Michael Ichikawa

AI ENGINEER · ML ENGINEER · DATA SCIENTIST

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SUMMARY

AI/ML engineer who builds AI agents and the multi-agent systems that coordinate them. As Founding AI/ML Engineer at Microclaw LLC, I designed and shipped a production AI agent for Microsoft 365, now live on the Microsoft Commercial Marketplace: a plan/act/observe loop over 85 tools with KNN tool retrieval, RAG semantic memory, and an in-house eval framework. Alongside it I built a multi-agent workflow engine from scratch: modular specialists composed into validated workflows with a measured learning loop. Underneath, a strong quantitative core: MS Mathematics, eight years teaching undergraduate math, and three years of semiconductor physical design at Intel.

FLAGSHIP WORK

Microclaw — Production AI Agent for Microsoft 365

Live on the Microsoft Marketplace

A single AI agent operating across 12 Microsoft 365 services through 85 function-calling tools. Iterative plan/act/observe orchestration loop (Azure OpenAI, function-calling), KNN tool retrieval (95.0% recall vs 92.7% baseline, benchmarked at 1,020 examples), RAG semantic memory, smart model routing, and multi-layer failure recovery. Designed, built, and shipped solo. Role detail under Experience.

TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep · 374 Vitest tests

Multi-Agent Workflow Engine — built from scratch

mtichikawa.github.io/workflow-engine

A from-scratch Python engine that turns a plain-English request into a working multi-agent workflow, built from a library of 14 modular, single-purpose agents with human approval before anything acts. It reuses existing agents instead of near-duplicating them, and I tested it against real public data across 12 unrelated domains, keeping the weaker results marked provisional rather than dropping them.

Python · agent orchestration · contracts + recipes · static graph validation · few-shot retrieval · LLM-as-judge

TECHNICAL SKILLS

Languages	Python · SQL · C++ · R · TypeScript · Node.js
ML / Statistics	scikit-learn · Prophet · PyTorch · XGBoost · LightGBM · scipy · statsmodels · Bayesian inference
LLM / Agents	agent orchestration · multi-agent workflows · MCP (Model Context Protocol) · KNN tool retrieval · RAG · Anthropic API · Azure OpenAI · Microsoft Graph · HuggingFace · FinBERT
Data Engineering	PostgreSQL · Databricks · Delta Lake · medallion · dbt · SQLAlchemy · pandas
Cloud / Infra	AWS (S3 · Lambda · DynamoDB · CloudWatch) · Docker · FastAPI · Redis · Pydantic v2 · boto3 · Microsoft Commercial Marketplace
Tools	Git · pytest · Vitest · Jupyter · Streamlit · Docker Compose

EXPERIENCE

Founding AI/ML Engineer

2026 – present

Microclaw LLC · Portland, OR

- Founding engineer: designed, built, and shipped the agent end to end to the Microsoft Commercial Marketplace (Standard + Self-Hosted offerings). Iterative plan/act/observe loop over 85 function-calling tools across 12 Microsoft 365 services, two-stage KNN + keyword tool selection, and multi-layer failure recovery (backoff/retry across the LLM and Graph APIs, loop detection)
- RAG semantic memory, a smart model router (GPT-4o-mini default, mid-task escalation to GPT-4o), and an in-house eval framework (107 tests, A/B vs a baseline bot) that gated releases; natural-language automations parsed to structured Postgres rules and fired via Microsoft Graph webhooks. 374 Vitest tests; Postgres row-level security for multi-tenant isolation; Bicep IaC; Docker → GHCR CI

Mathematics Faculty
2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for eight years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Designed assessments and iterated course materials from cohort performance data
- Taught STEM majors alongside non-STEM majors fulfilling general-education math requirements at an open-enrollment community college

Physical Design Engineer
2008 – 2011

Intel Corporation · Semiconductor R&D

- Microprocessor physical design on Intel Ivy Bridge (22nm), including timing closure, power analysis, and design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run

EDUCATION

MS Mathematics	The City College of New York	2014
BS Mechanical Engineering	University of California, Berkeley	2008

PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Beyond the two flagships above, a body of self-directed work across the data and ML stack.

Trading System Arc — five-repo paper trading system: ccxt/Kraken OHLCV → mplfinance charts → EMA/RSI/MACD + FinBERT signal fusion → parameter-optimizing backtester → Streamlit oversight dashboard. 174 tests. live demo

Databricks Medallion Lakehouse — bronze/silver/gold on 10M NYC taxi rows, Delta Lake, row-level data-quality gates between layers. 95 tests.

GCP RAG Pipeline — production-shape RAG: Vertex AI Gemini + BigQuery Vector Search, LangChain/LlamaIndex, six-combination retrieval benchmark on Cloud Run.

Real-Time Anomaly Detection — streaming Isolation Forest + LSTM + Z-score ensemble, Dockerized FastAPI, live interactive demo.

Cloud ETL Pipeline — AWS S3/Lambda/DynamoDB, Parquet + Hive partitioning, moto-tested.

Also: SQL Analytics Pipeline · LLM Data Assistant (hybrid routing) · Multi-Armed Bandit A/B Testing (Thompson Sampling · UCB1) · LLM Bias Detection (ANOVA · Cohen's d) · Financial NLP Parser · GitHub Trend Forecaster (Prophet) · Streaming Analytics Pipeline (Spark/Redpanda) · C++/pybind11 rolling stats.